

## 6.7 Presenting Data Using Digital Tools

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔

CS XI

Chapter 6: Digital Literacy

Lecture 08 of 14

### 6.7 Presenting Data Using Digital Tools

#### LEARN IT

Collecting data is only **one part** of the work. You must also **present** your findings so that others can understand them **quickly and easily**. Digital tools make this much easier. The four main tools are spreadsheets, presentations, infographics and reports.

#### 6.7.1 Spreadsheets

#### LEARN IT

**Spreadsheets** such as **MS Excel**, **Google Sheets** and **Apple Numbers** arrange information in **rows** (horizontal, numbered 1, 2, 3...) and **columns** (vertical, lettered A, B, C...). The box where a row and column meet is a **cell**, named by its column and row, e.g. **B2**. With a spreadsheet you can:

- **Enter data** systematically (one record per row).
- **Calculate** totals, averages and percentages.
- **Sort** (arrange in order) and **filter** (show only some rows) records.
- **Create charts and graphs** automatically.

A **formula** always starts with =. A **range** such as B2:B4 means "cells B2 to B4".

| Formula                   | What it does                                       |
|---------------------------|----------------------------------------------------|
| =SUM(B2:B4)               | Adds all values in B2 to B4.                       |
| =AVERAGE(B2:B4)           | Mean: sum ÷ number of values.                      |
| =MAX(B2:B4) / =MIN(B2:B4) | Largest / smallest value.                          |
| =MEDIAN(B2:B4)            | Middle value when sorted.                          |
| =COUNT(B2:B4)             | Counts cells containing numbers.                   |
| =COUNTA(C2:C4)            | Counts cells that are not empty (text or numbers). |
| =COUNTIF(C2:C4, "Yes")    | Counts cells that match a condition.               |

### WORKED EXAMPLE

**Analysing the case-study spreadsheet.** The data from the 6.8 case study (Step 8) is entered like this:

|   | A       | B     | C         | D             | E           |
|---|---------|-------|-----------|---------------|-------------|
| 1 | Student | Hours | Study Use | Main Activity | Distracted? |
| 2 | 1       | 3     | Yes       | Lectures      | Sometimes   |
| 3 | 2       | 5     | No        | Social Media  | Yes         |
| 4 | 3       | 2     | Yes       | Notes         | No          |

**Step 1 – Total hours:** =SUM(B2:B4) → 3 + 5 + 2 = **10**.

**Step 2 – Average hours:** =AVERAGE(B2:B4) → 10 ÷ 3 = **3.33** hours (to 2 decimal places).

**Step 3 – Most and least:** =MAX(B2:B4) → **5**; =MIN(B2:B4) → **2**.

**Step 4 – Median:** sorted values 2, 3, 5 → middle = **3**. =MEDIAN(B2:B4) gives 3.

**Step 5 – How many use the phone for study?** =COUNTIF(C2:C4, "Yes") → **2**.

**Step 6 – Percentage:** =COUNTIF(C2:C4, "Yes")/COUNTA(C2:C4) → 2 ÷ 3 = 0.667; format the cell as % → **66.7%**.

**Step 7 – How many said "Yes" to distracted?** =COUNTIF(E2:E4, "Yes") → **1** (only "Yes" matches; "Sometimes" does not).

**Step 8 – Sort:** select the data → Data → Sort by column B (largest to smallest) → order becomes Student 2 (5), Student 1 (3), Student 3 (2).

## 6.7.2 Presentations

### LEARN IT

Presentation software such as **PowerPoint**, **Google Slides** and **Canva** helps you explain findings to an **audience** in an organised way. Good practice:

- Begin with a clear **title** and the **research question**.
- Use **short bullet points** instead of long paragraphs.
- Include **tables, charts and key findings**.
- Keep the design **simple and readable** (large font, few colours).

Spreadsheets and presentations work together: you make a chart from the spreadsheet data and insert it into the slides.

Spreadsheet data



Chart / Graph



Insert



Slides

### REMEMBER IT

Presentation rule of thumb: **one idea per slide**, and no more than about **six short bullet points**. If you are reading long sentences off the slide, the slide has too much text.

### COMMON MISTAKE

**Mistake:** Typing a result by hand (e.g. writing "3.33" in a cell) instead of using a formula. If the data changes, the typed result becomes wrong. **Fix:** Always use formulas like `=AVERAGE(B2:B4)` so results update automatically. Also remember every formula must begin with `=`.

### اردو میں سمجھیں

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔ اسپریڈشیٹ (جیسے ایکسل اور گوگل شیٹس) میں ڈیٹا قطاروں اور کالموں میں ترتیب دیا جاتا ہے، اور `=SUM()` اور `=AVERAGE()` جیسے فارمولوں سے مجموعہ، اوسط اور فیصد نکالے جاتے ہیں۔ پریزنٹیشن سافٹ ویئر (جیسے پاور پوائنٹ) میں مختصر نکات، چارٹس اور سادہ ڈیزائن کے ساتھ نتائج سامعین کے سامنے پیش کیے جاتے ہیں۔

### CHECK YOURSELF

1. Write the formula to find the average of cells D2 to D20.
2. Name three spreadsheet programs.
3. Give two rules for a good presentation.

Answers: 1. `=AVERAGE(D2:D20)` . 2. MS Excel, Google Sheets, Apple Numbers. 3. Any two: start with a clear title and research question; use short bullet points; include tables/charts/key findings; keep design simple and readable.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Spreadsheets are good for:

- a. Tables, calculations, simple charts
- b. 3D games
- c. NAND simulation
- d. OSI sniffing

**MCQ 2.** A presentation (slides) is for:

- a. Talking an audience through findings
- b. Storing the only copy of raw data
- c. Primary keys
- d. Kernel drivers

**MCQ 3.** An infographic is:

- a. A visual one-pager of facts
- b. A 40-page appendix
- c. A binary search
- d. A foreign key

**MCQ 4.** A report is:

- a. Structured writing with method, results, limits
- b. Only memes
- c. Only code
- d. Only ER

**MCQ 5.** Choose the tool by:

- a. Audience and purpose
- b. Whatever is newest
- c. Always infographic
- d. Always 80 slides

**MCQ 6.** Raw 2000-row sheets in a 5-minute talk are:

- a. A bad present
- b. Ideal
- c. Required by BIEK
- d. A sort algorithm

## Extra short questions

1. Four digital presentation tools in the book.
2. When spreadsheet > slides.
3. When infographic > report.
4. One rule for all four.

## Extra long question

1. Same finding (70% of surveyed students want longer labs). Design: a spreadsheet cell, a slide headline, an infographic hook, and a report paragraph.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Spreadsheets, slides, infographics, reports.
2. When the audience must check the numbers or filter.
3. Awareness poster / social post — not a methods thesis.
4. Cite sources; label axes; do not crop axes to lie.

**Long**

1. Sheet: table+bar. Slide: one number, one chart, one ask. Infographic: 7 of 10 icon row.  
Report: sample n, question wording, 70%, limit (one college).

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Next lecture: 6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports